Machine Learning Project

Image Segmentation

What is segmentation?

Image segmentation assigns a class label to every pixel of an image. The output is a mask of the same height and width as the input. In these projects each pixel is either the object or the background. The predicted mask is compared with the human-drawn mask. Precision is the fraction of predicted object pixels that are correct. Recall is the fraction of true object pixels that were found. The F1 score combines precision and recall. Intersection over Union (IoU) is the area of overlap divided by the area covered by either mask. The result is better when these scores rise on images that were not used for training.

A compound loss is a training objective made by adding two or more loss terms. One term can score each pixel, and another term can score the overlap of the whole predicted mask with the true mask.

Each project below is a small binary task. Resize every image to 256 by 256. Make an appropriate train, validation, and test split for the size of the dataset. Use the same split for the baseline and the proposed model. On the test split, report precision, recall, F1, and IoU for both models.

1. Lesion segmentation

Lesion segmentation marks the diseased region in a skin photograph. The lesion is one connected region. Cross-entropy scores each pixel on its own, so the border, which covers only a small share of the pixels, can remain blotchy.

PH2 contains 200 dermoscopic photographs, each 768 by 560, and each photograph has a lesion mask.

Dataset: https://www.kaggle.com/datasets/spacesurfer/ph2-dataset

Problem statement 1. Segment the lesion with a U-Net. Use cross-entropy as the baseline. Explore compound losses, select one, write its formula, and state why that loss was chosen. Train the same U-Net with the selected loss. The compound loss should improve the test scores.

2. Crack segmentation

Crack segmentation marks a thin crack in a pavement photograph. Almost every pixel is background, so cross-entropy spends most of its effort on the pavement. The crack then comes out broken or too thick.

CrackForest (CFD) contains 118 colour pavement photographs of about 480 by 320 pixels, and each photograph has a crack mask.

Dataset: https://github.com/cuilimeng/CrackForest-dataset

Problem statement 2. Segment the cracks with a U-Net. Compare ordinary cross-entropy with a weighted cross-entropy that counts crack pixels more heavily. Write the weighted loss and choose the weight on the validation split. In one ablation, set the weight back to 1. The weighted loss should improve the test scores.

3. Horse segmentation

Horse segmentation marks the horse in a side-view photograph. The background may be grass, a fence, or a person, so colour alone is not a reliable rule. Shrinking the image inside the encoder can drop the outline. Skip connections carry that detail back to the decoder.

The Weizmann Horse set contains 327 side-view photographs, each with a binary horse mask.

Dataset: https://www.kaggle.com/datasets/ztaihong/weizmann-horse-database

Problem statement 3. Segment the horse, using every training mask. Choose one loss and use it for both models. Compare an encoder-decoder that has no skip connections with the same network after those connections are restored. The skip connections should improve the test scores. Show two test photographs where the outline is better.

4. Person segmentation

Person segmentation marks every person in a street or campus photograph as one foreground class, including people who overlap. The collection is small. People face left or right and stand at different distances, so training on the photographs unchanged can memorise those few views.

The Penn-Fudan Pedestrian set contains 170 photographs and 345 person masks.

Dataset: https://www.cis.upenn.edu/~jshi/ped_html/PennFudanPed.zip

Problem statement 4. Merge all people in a photograph into one binary mask. Do not separate people who overlap. Train one U-Net with no augmentation and the same U-Net with a horizontal flip and a small zoom. Use the same loss for both. Test both models on the original photographs. Augmentation should improve the test scores.

Submission Link: https://forms.gle/ufJzdeXSwctMPfFq9

Submit the link to a folder named MLproject_rollno_topic. The folder must contain the code, the slides, and the report.

Project Evaluation Basis:

S. No.	Evaluation Criteria	Evaluation Parameters	Max. Marks
1	Title, introduction and contribution	Problem clarity, motivation, and the change studied in the chosen segmentation project	5
2	Literature review	Existing segmentation methods, their limitations, and the reason for the proposed change	5
3	Dataset and preprocessing	Dataset check, resize to 256 by 256, and a train, validation, and test split suited to the dataset size	5
4	Implementation of the baseline and the proposed model	Correct training of both models, the same split and settings, and hyperparameter tuning	15
5	Proposed methodology, comparison with the baseline, and ablation	The explored change, its formula or rule, and the comparison when that change is removed	10
6	Results, discussion, and comparative analysis	Precision, recall, F1, and IoU in a table and a graph, with interpretation of the test masks	10
7	Technical viva and code demo	Understanding of the model, the change, the loss, and the segmentation metrics	30
8	Presentation (10 slides) and documentation (8 pages maximum)	Slide content, architecture diagram, code, and report	20
		TOTAL	**100**

Project status and progress discussion and presentation dates: 3rd, 10th and 17th November 2026.